FARM MANAGEMENT SYSTEM

Project work submitted to Thiruvalluvar University, Serkkadu, Vellore

In Partial Fulfillment For The Award Of Degree

BACHELOR OF SCIENCE

IN

COMPUTER SCIENCE

By

Team Leader : NAVEENKUMAR G - 20923U18129
Team member : NOEL ROSHAN J – 20923U19131
Team member : PANDHIYAN B – 20923U18132
Team member : PANNEERSELVAM P - 20923U18133
Team member : POONGAVANAM R – 20923U18135

PG & RESEARCH DEPARTMENT OF COMPUTER SCIENCE

SHANMUGA INDUSTRIES ARTS &SCIENCE COLLEGE

Accredited with "B+" Grade by NAAC,
Certified Under Section 2(f) & 12B of the UGC Act 1956,
An ISO 9001:2015 Certified Institution
Permanently Affiliated to Thiruvalluvar University, Vellore,
Approved by Govt. of Tamilnadu & AICTE,
Tiruvannamalai – 606603.

MARCH – 2026

1. INTRODUCTION

1.1 Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose :

The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient

PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

2.2 Empathy Map Canvas

Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

2.3 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

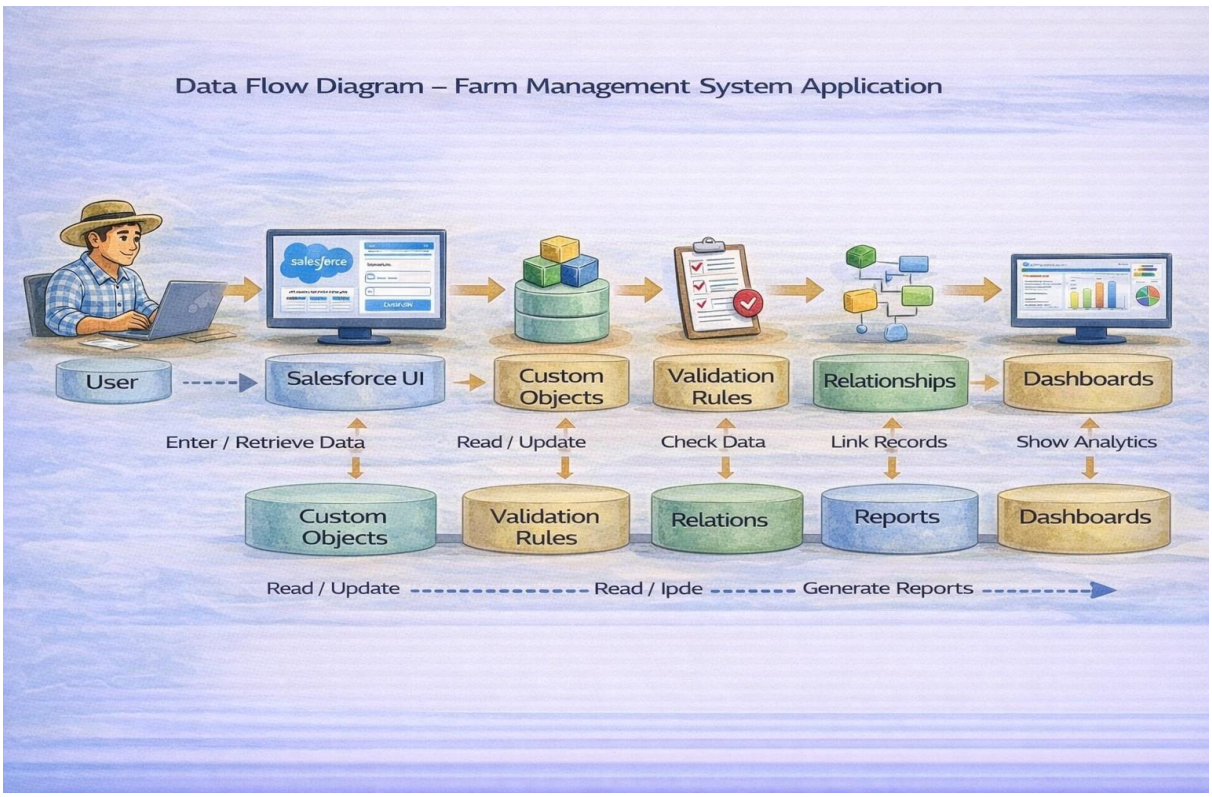
1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results
7. Decision Making

3.2 Solution Requirements
Functional Requirements

FR No	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation
FR-6	Dashboard Visualization

3.3 Data Flow Diagram

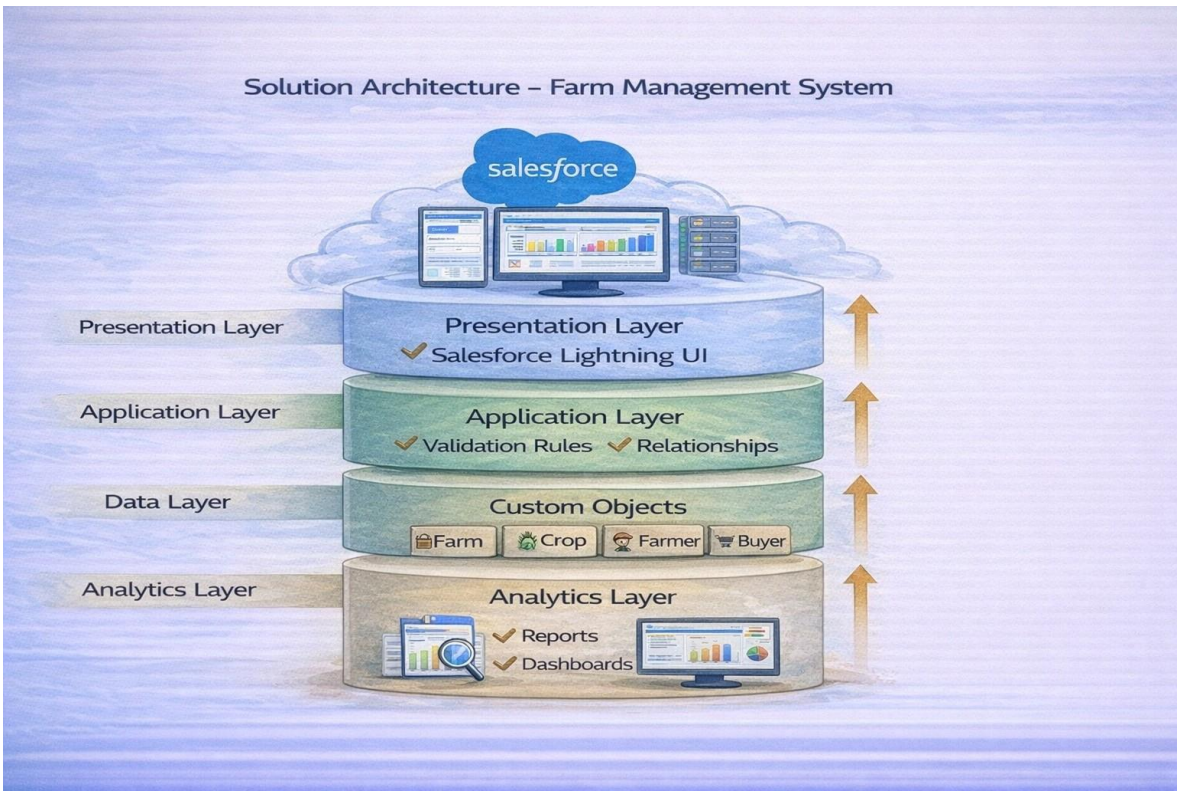
Data Flow:
User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards



Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

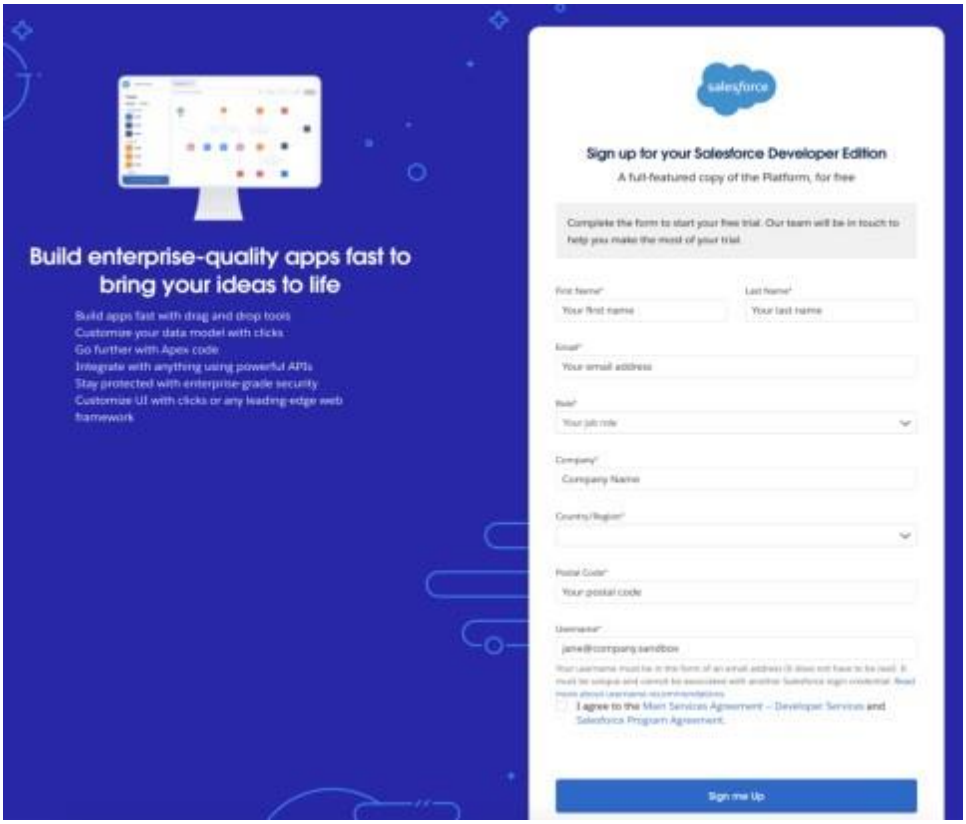
Activity1:

Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search Developer.salesforce.com/signup
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.

4. Click save.
5. Search login.salesforce.com



6. By using username and password you can into the Salesforce org. The homepage will appear below

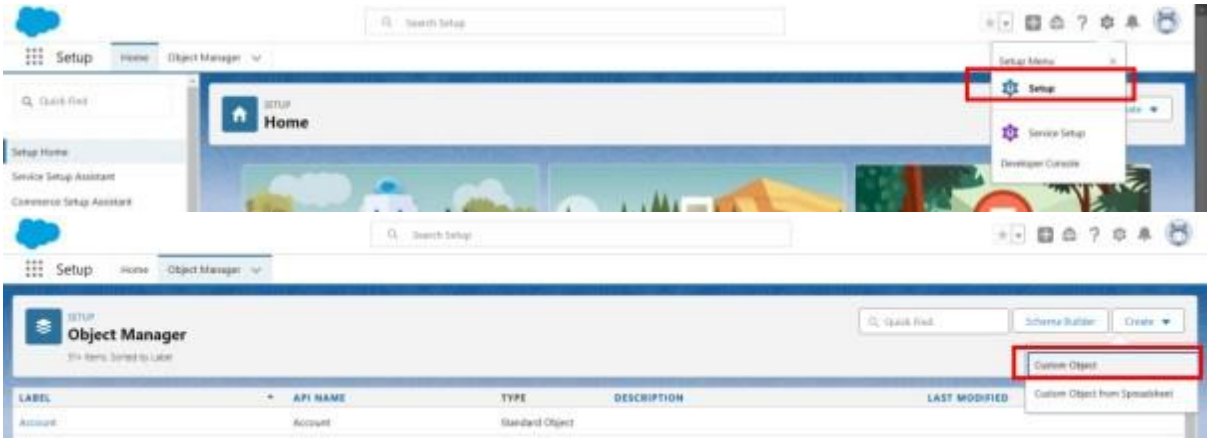
Milestone 2-Object:

What is an object?
Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:
1.Farm
2.Crop
3.Farmer
4.Buyer

Activity 1:
Create a custom object for Farm:
To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as " Farm ".
4. Fill in the plural label as " Farms".
5. Record name: Farm Name"
6. Select the data type as "Text".
7. In the Optional Features section, select Allow Reports and Track Field History.
8. In the Deployment Status section, ensure Deployed is selected.
9. In the Search Status section, select Allow Search.
10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.

a. On the Custom Object Definition page, create the object as follows: b. Label: Crop

c. Plural Label: Crops
d. Record Name: Crop Name

e. Select the data type as "Text".
f. Check the Allow Reports checkbox g. Check the Allow Search checkbox

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout i. Click Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.

a. On the Custom Object Definition page, create the object as follows:
b. Label: Farmer
c. Plural Label: Farmers
d. Record Name: Farmer Name e. Select the data type as "Text".

f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Activity 4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.

a. On the Custom Object Definition page, create the object as follows:
b. Label: Buyer
c. Plural Label: Buyers
d. Record Name: Buyer Name
e. Select the data type as "Text".
f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Milestone 3 -
Tab: Tab

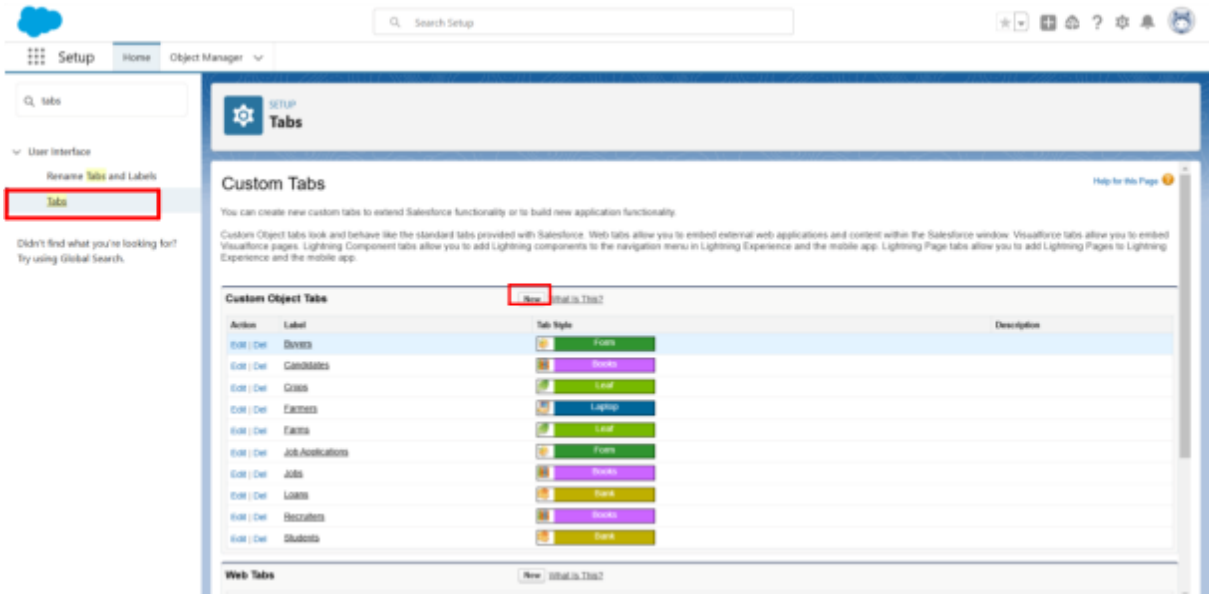
Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.



2. For Object, select Farm.
3. For Tab Style, select any icon.
4. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.

4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

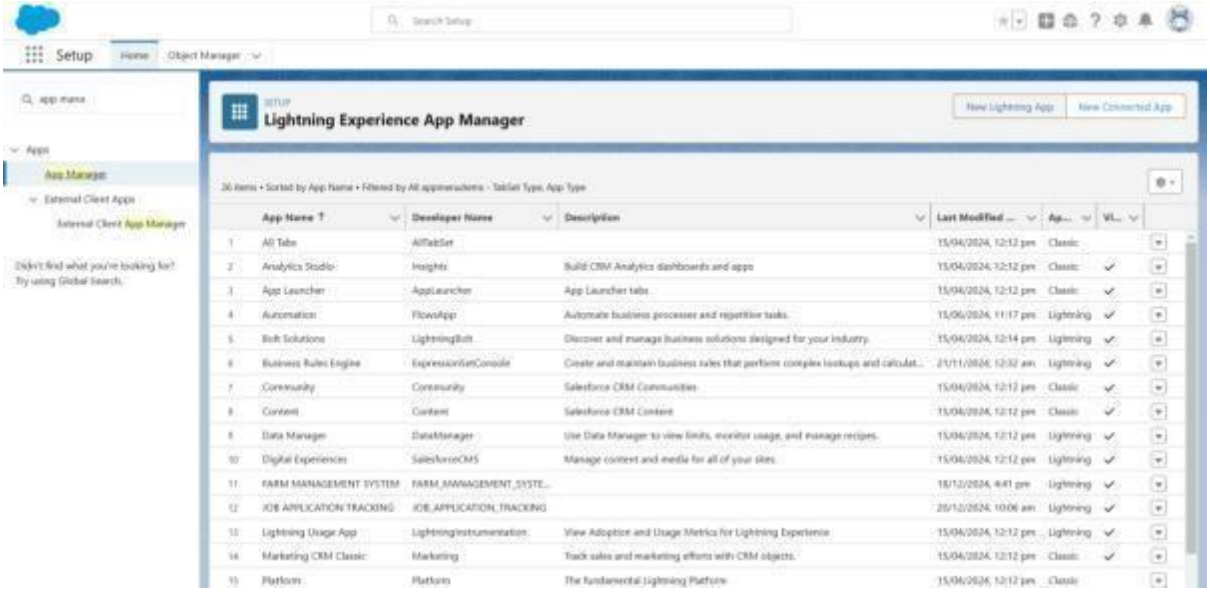
Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce area group of tabs that help the application function by working together asa unit. It has a name,a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

Create the Farm Management System app



- From Setup, enter App Manager in the Quick Find and select App Manager.

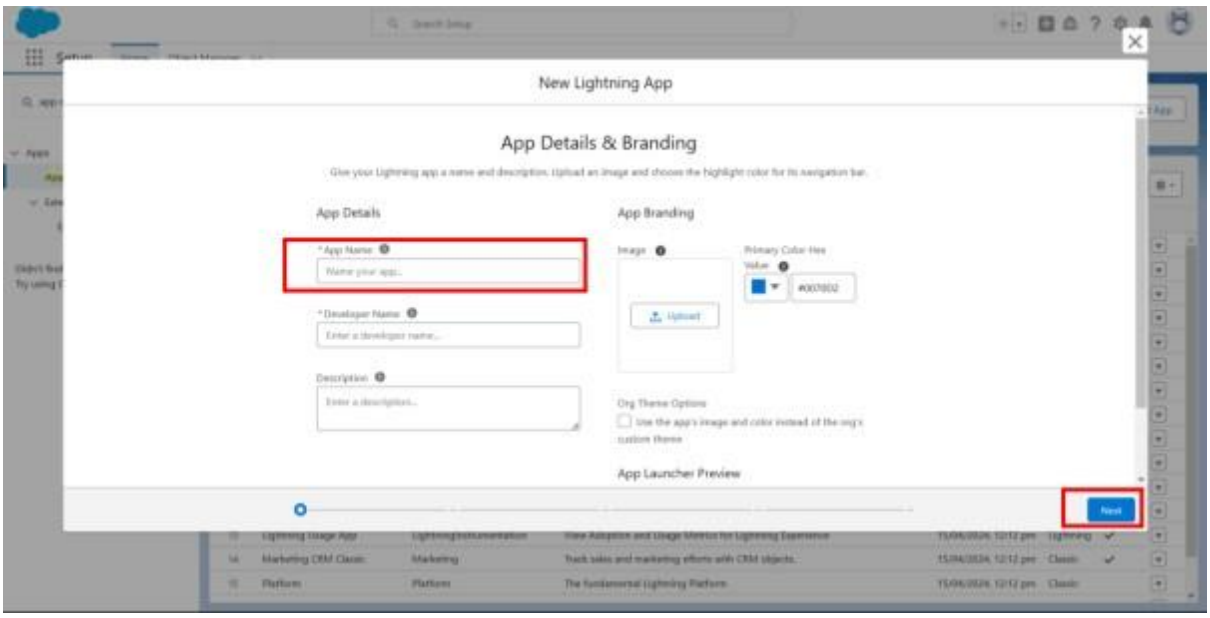
- Click New Lightning App. Farm Management System as the App Name, then click Next
- Under App Options, leave the default selections and click Next.
- Under Utility Items, leave as is and click Next.
- From Available Items, select Farm, Crop,Farmer,Buyer

Reports,and Dashboards and move them to Selected Items. Click Next. ● From Available Profiles, select System Administrator and move it to Selected Profiles.Click Save & Finish.

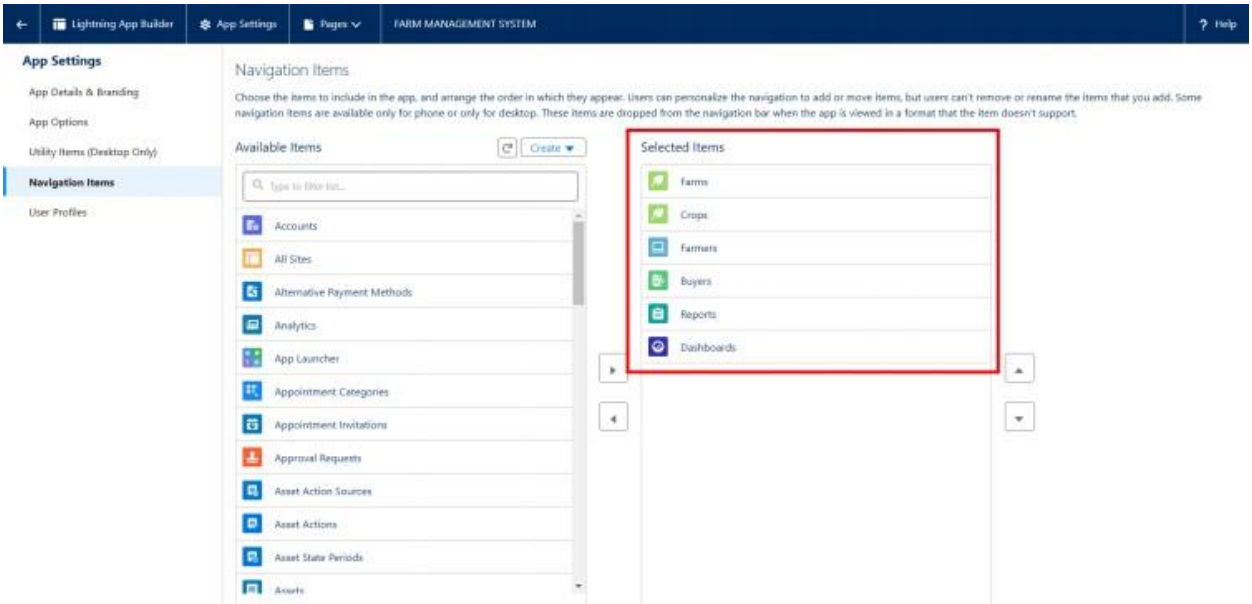
- To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

Note:

1. App Launcher-Displays available apps.
2. App Name-Displays the current selected app.



3. Navigation menu-Displays the tabs available inside the app.

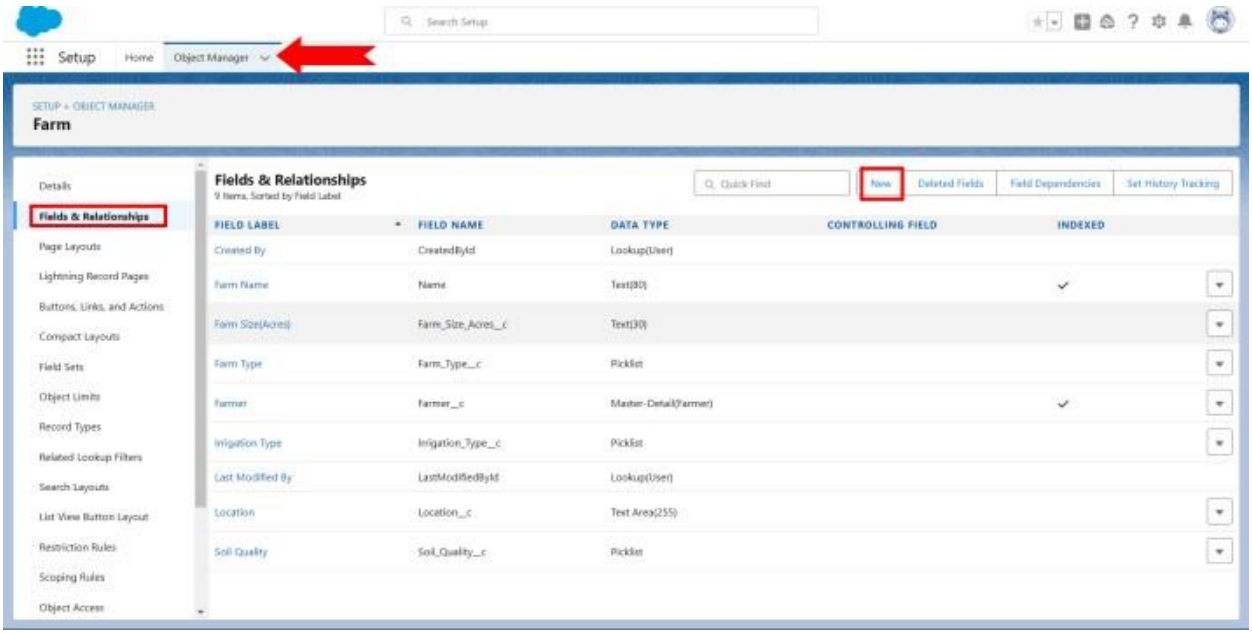


Milestone 5 -
Fields and Relationship: What are fields?

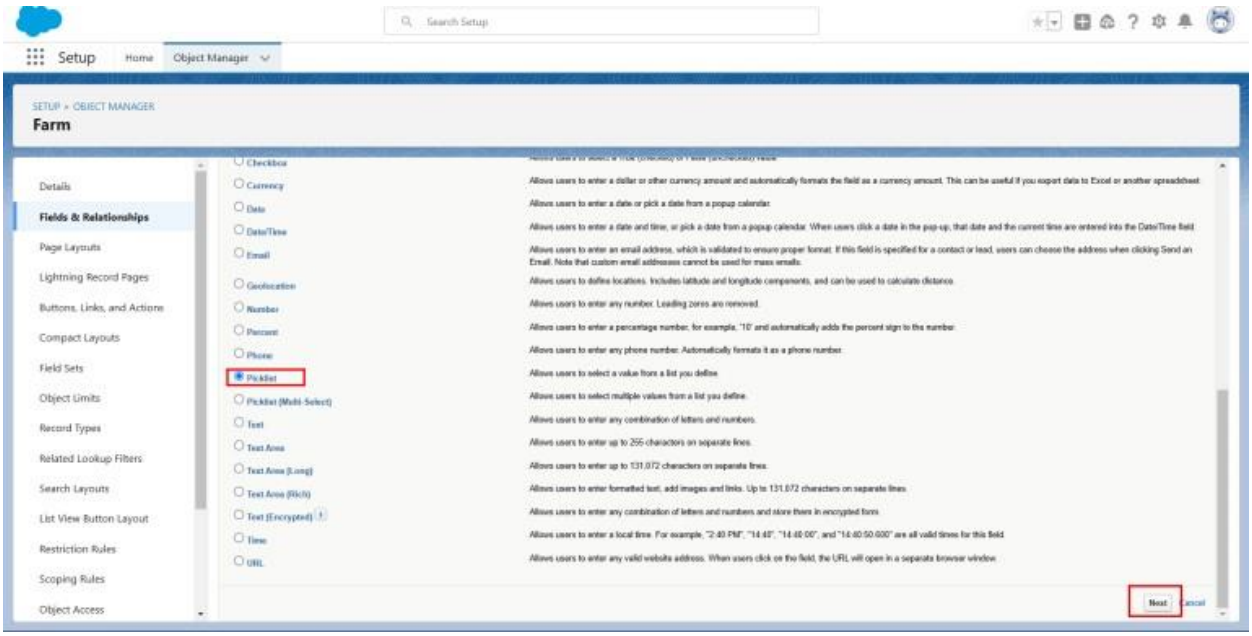
Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1:
Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New

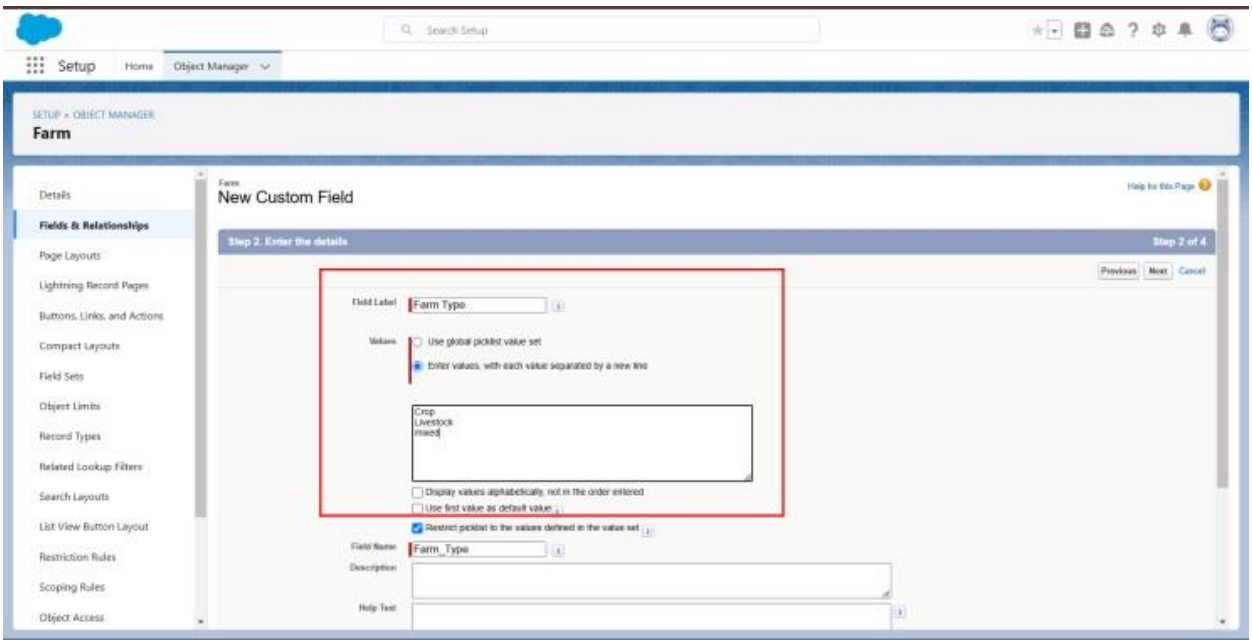


5.Choose Data type as a picklist,Click next



6.Enter Field Label (Farm Type),and Enter Values as (Crop,Livestock,Mixed)

7.Next,Next Click Save.



Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)
2.Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)
3.Choose data type as a text,enter field label(Location) 4.Choose data type as text,enter field label (size)
5.Create a Master Detail relationShip with Farmer Object.

Activity 2:
Creation of fields for the Crop objects:

1.Choose Data type As a text,Enter Field Label name (Crop Name),Next save. 2.Select Data type as date ,Enter Field label (Planting Date),then save .
3.Again we select data type as a date,Enter field labe (Harvest date),save
4.Next We Create Lookup relationship on crop object realted to Farm Object.
5.We select adata type as currency,Enter field Label (Actual Yield), Next Save it. 6.Select Data type as a Number,enter Field Label (Total Acres Harvested),Enter Maximum Length(20) and save.
7.Now We create Formula Field For That Select data Type As a Formula,Click Next.
.Enter Field Label (Expected Yield per Acre),Select formula return type as Currency,click Next.
9.Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.

Activity 3:
Creation of fields for the Farmer objects:

1.Select Data type Phone,Enter field label(Contact Number),next Save.
2.Select Data type Picklist,Enter Field Label(State),Enter values (Telangana,Andhra Pradesh,Tamilnadu),save.
3.Choose Data type Picklist,Enter Field Label (District),Enter Values of district,next save. 4.Again select Data type as Picklist,Enter Field Label (Village),Enter Values of villages,next save.

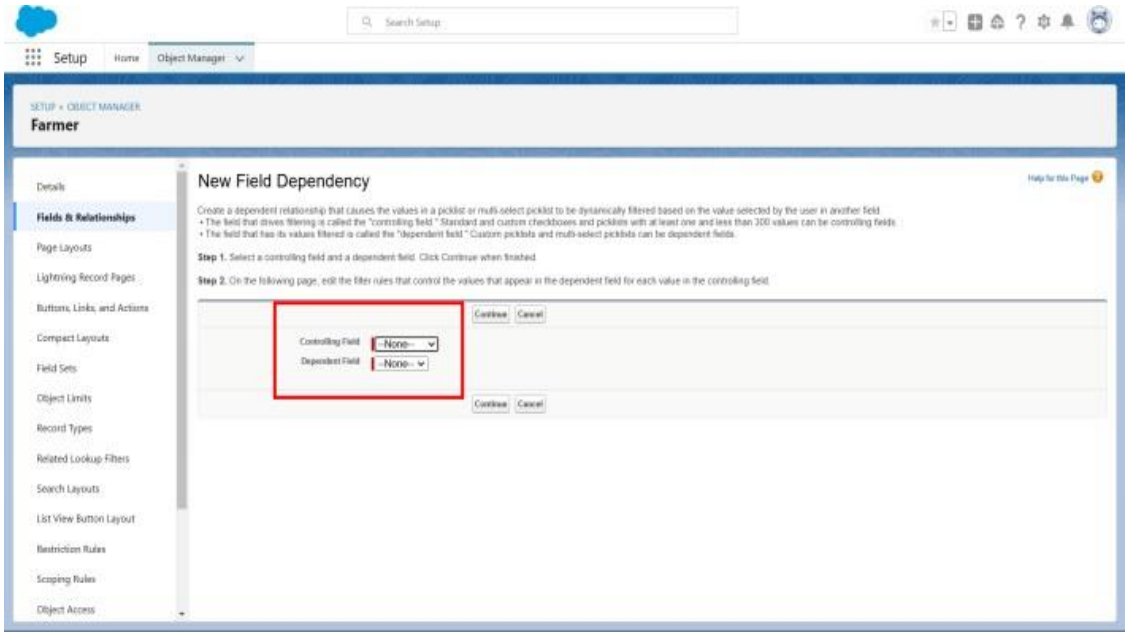
Creation of field dependency :

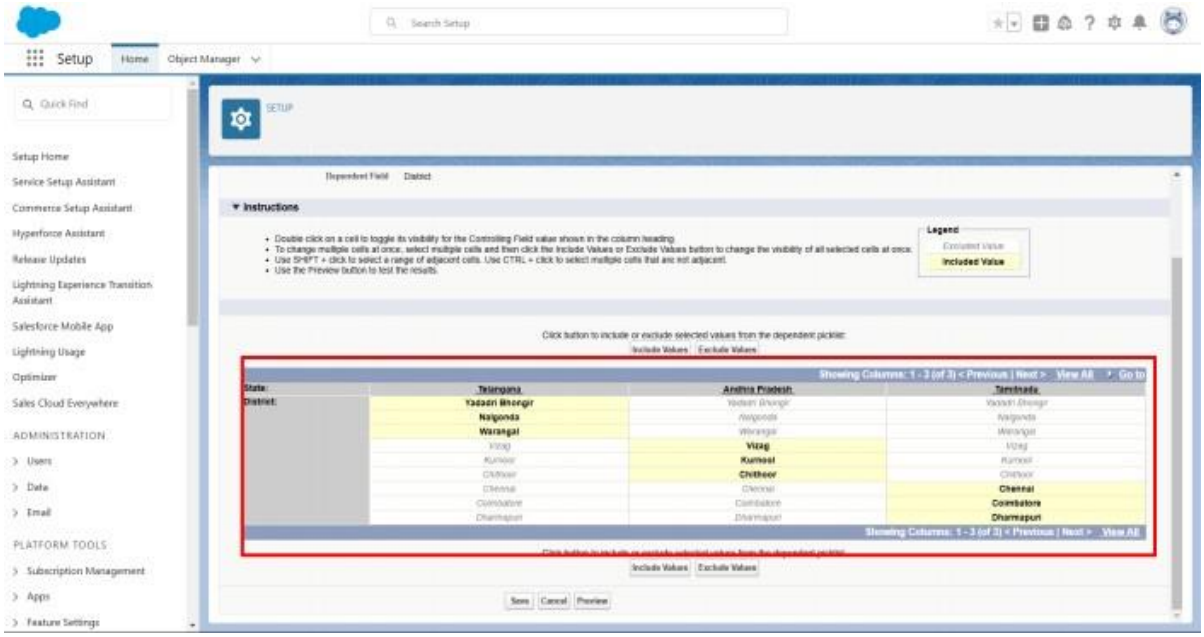
1.Now create Field Dependency For District field,Open Object Manager,Choose Farmer object ,select Fields&Relationships,On Right side We have Field Dependency option,select that option.
2.Click new, Select Controlling Field as State,Dependent Field As District,Click on Continue.

3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.

5.Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.

6.Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.





7. Click Close to close the preview window, Click Save.

Activity 4:

Creation of fields for the Buyer objects:

1.Select Data type as a Phone,Enter a Field Label(Contact),Save.
2.Select Data type as an Email,Enter Field Label(Email), Next save.
3.Select Data type Text area,Enter Field Label(Address),Save it.
4.Select Data type Picklist, Field Label(Buyer Type),Enter values (Wholesale,Retail),Click Save.

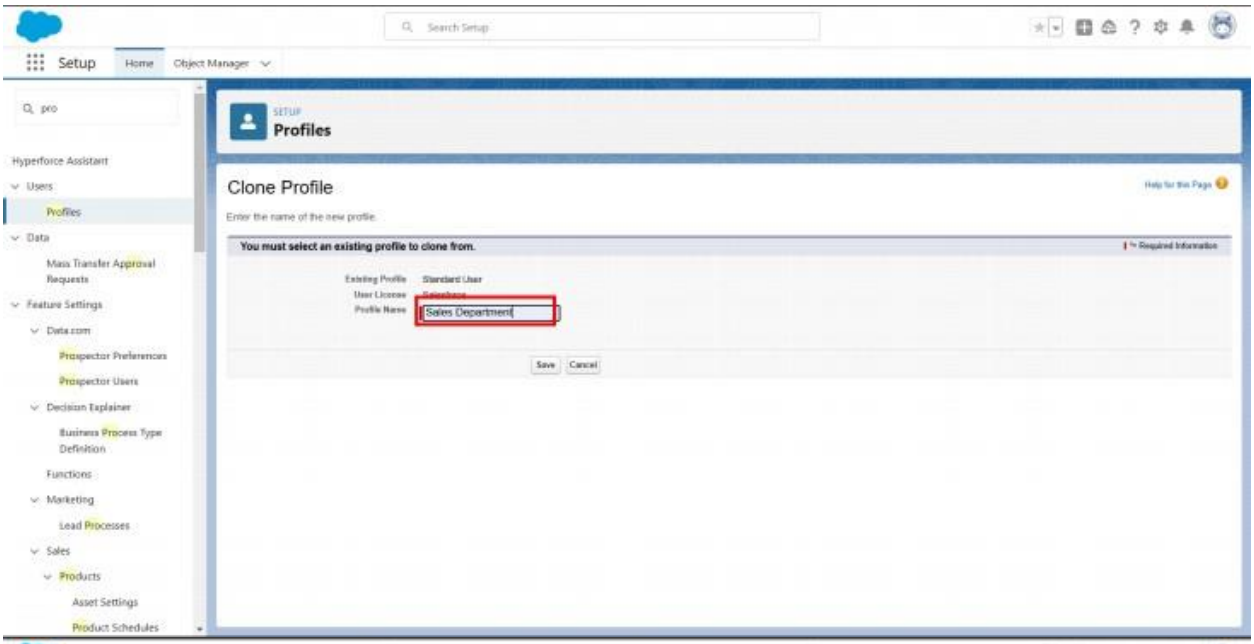
Milestone 6- Profile:

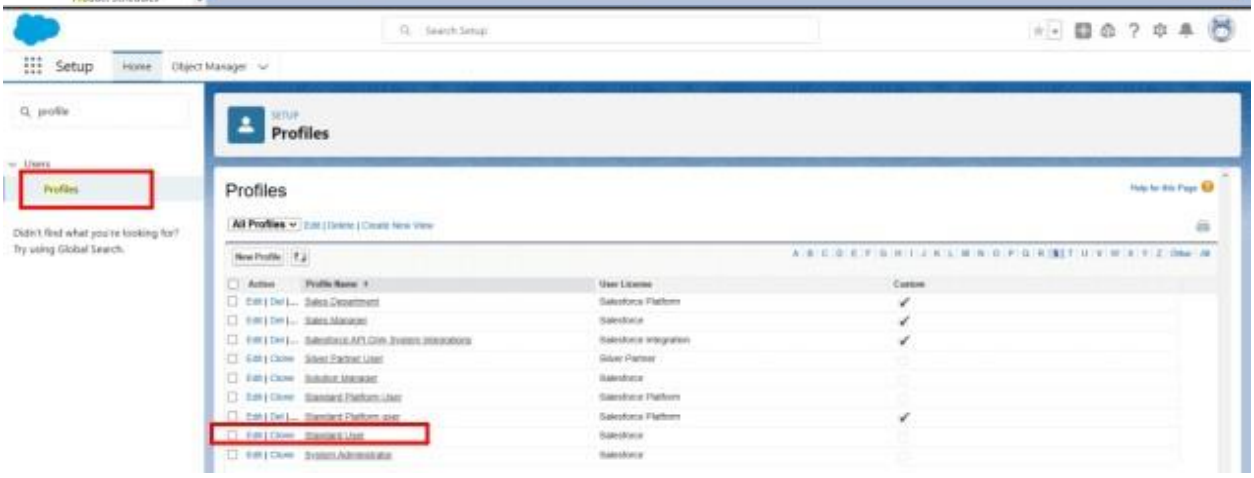
What is a profile?
A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls "Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.
A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.


Search Setup
Setup
Home
Object Manager
pro
Hyperforce Assistant
Users
Profiles
Data
Mass Transfer Approval Requests
Feature Settings
Data.com
Prospector Preferences
Prospector Users
Decision Explainer
Business Process Type Definition
Functions
Marketing
Lead Processes
Sales
Products
Asset Settings
Product Schedules
SETUP
Profiles
Clone Profile
Help for this Page
Enter the name of the new profile.
You must select an existing profile to clone from.
Required Information
Existing Profile
Standard User
User License
Salesforce
Profile Name
Sales Department
Save
Cancel


Search Setup
Setup
Home
Object Manager
profile
Users
Profiles
Didn't find what you're looking for?
Try using Global Search.
SETUP
Profiles
Profiles
Help for this Page
All Profiles
Edit | Delete | Create New View
New Profile
Action
Profile Name
User License
Custom
Sales Department
Salesforce Platform
Sales Manager
Salesforce
Salesforce API Only System Integrations
Salesforce Integration
Silver Partner User
Silver Partner
Solution Manager
Salesforce
Standard Platform User
Salesforce Platform
Standard Platform User
Salesforce Platform
Standard User
Salesforce
System Administrator
Salesforce

5. Click save.

Activity 2:

Create a custom profile-2:
Create a profile with the profile name as “ Farm Manager” .

6. From setup , enter profiles in Quick Find box
7. Select profiles (Standard user).
8. Click clone.

Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Activity-1
Creation of role

1. From the Quick find box search for the role and click on the roles option 2. select the set up roles option

3.Below the CEO click on add role and enter the label name as a ” Farm Manager ” and role name will be Automatically populated and click on save.

Milestone 8-User :

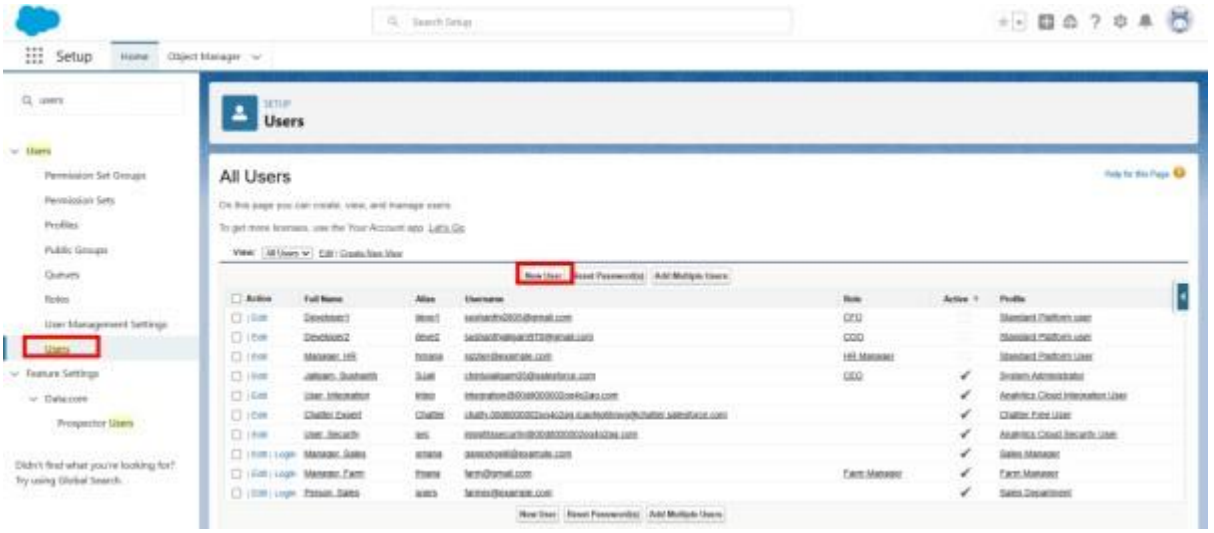
What is a user?
A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don’t find salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.



3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

Activity 2:
Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?
Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.
Types of sharing rules,
•Owner-based Sharing Rules •Criteria-based Sharing Rules

Activity 1:

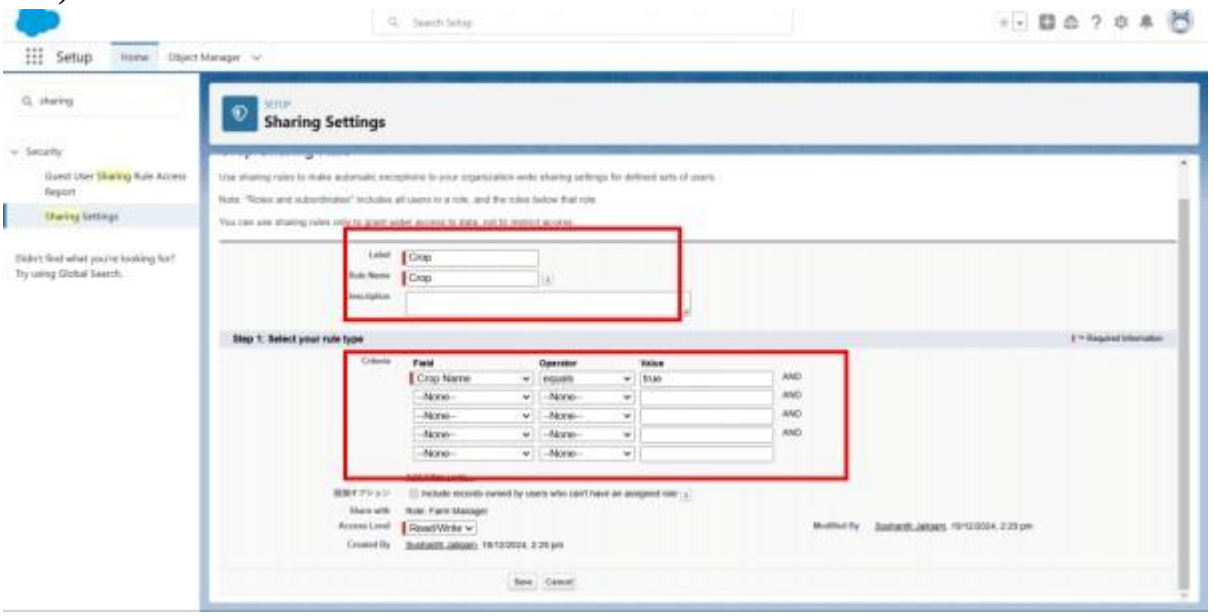
Create a sharing rule

1) Goto Sharing Settings, which can be found under the Quick Find section.
2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
3) Add the label of the sharing rule you want to make.
4) Select your rule type based on the criteria.
5) Select the field can join immediately check field from the Crop object.
6) Select the Crop as equal and value is True.

7) And in selecting the users to share with the section select roles and in that select Farm Manager.

8) And in the section of select the level of access for the users give the access Read/Write.

9) And save the rule.



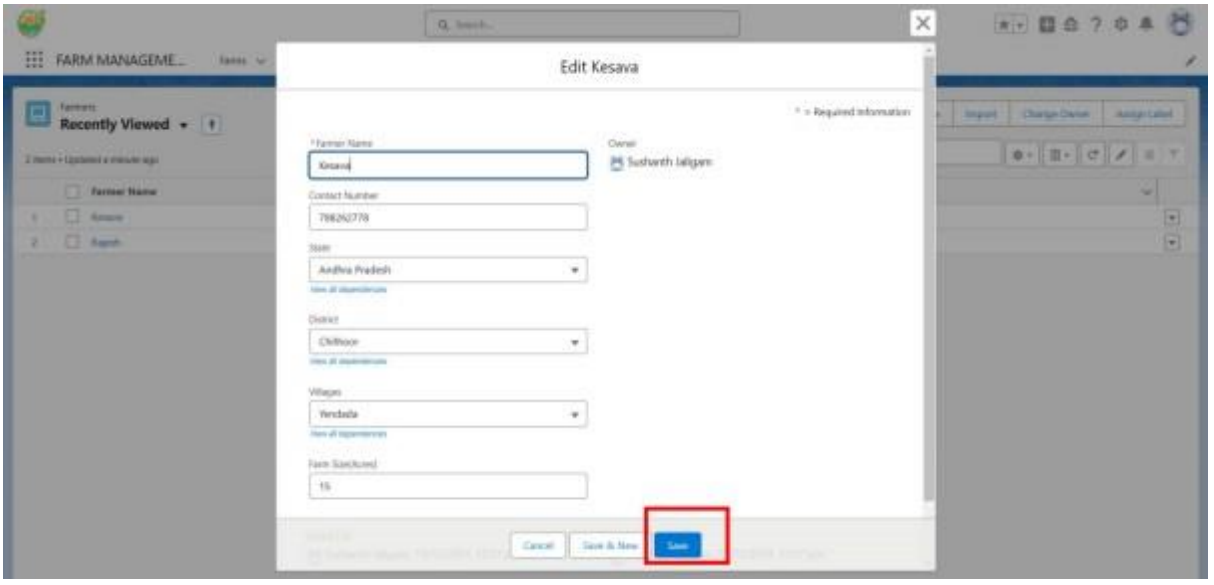
Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.
Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1:Create a Record(Farmer) ● Click on App Launcher on left side of screen.

- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.



Milestone 11-Reports:

What are Reports?

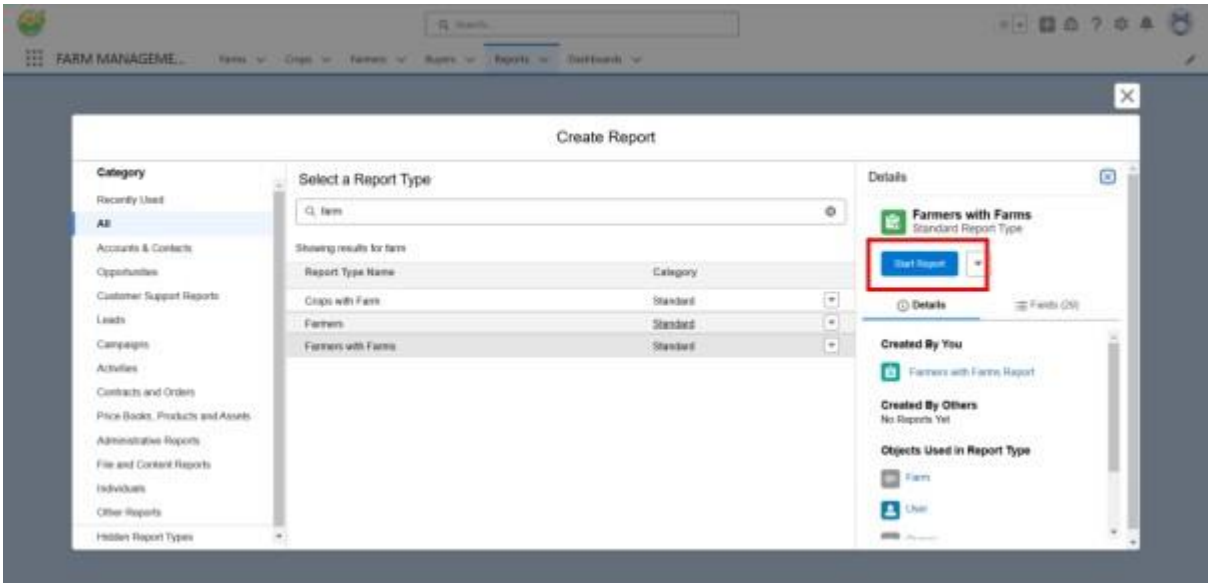
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

Create a report:

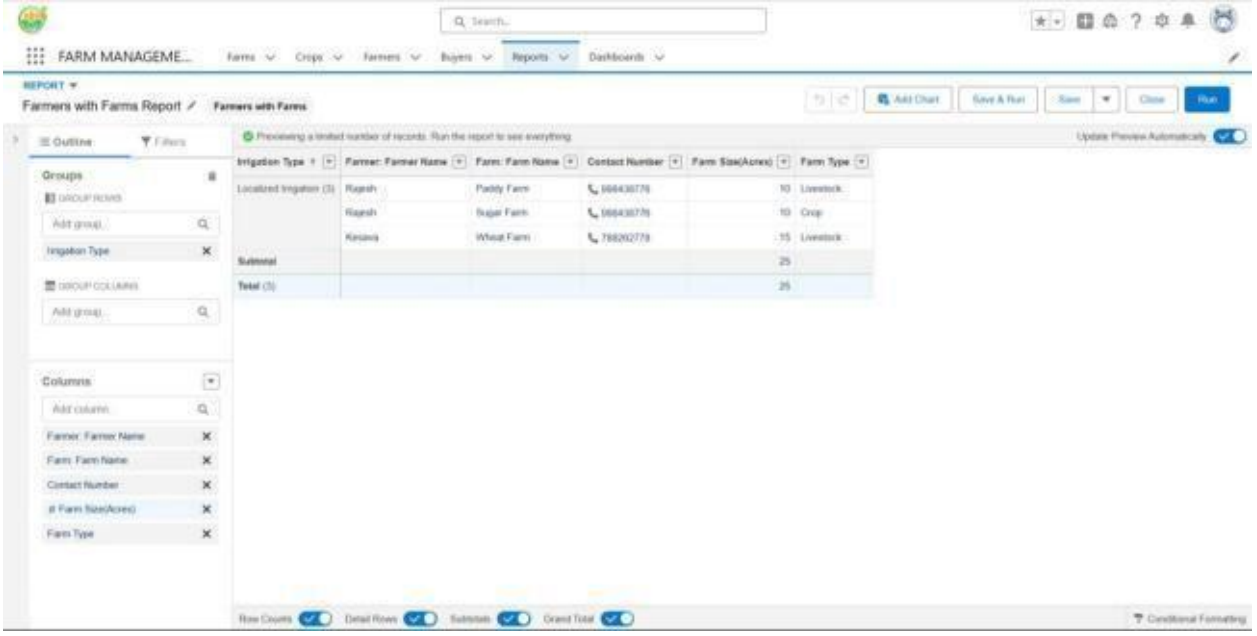
Create a report that displays rating of the account and which has type and account name.

1) Click on app launcher search for reports.
2) Click on the new report and select the category has Farmer with Farms.



3) In the details section select the option start report.
4) In Outline Click on Column Select Which Field you want to display in the Report.
5) Click Group,Select Which Fields You Want to Group.

6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.



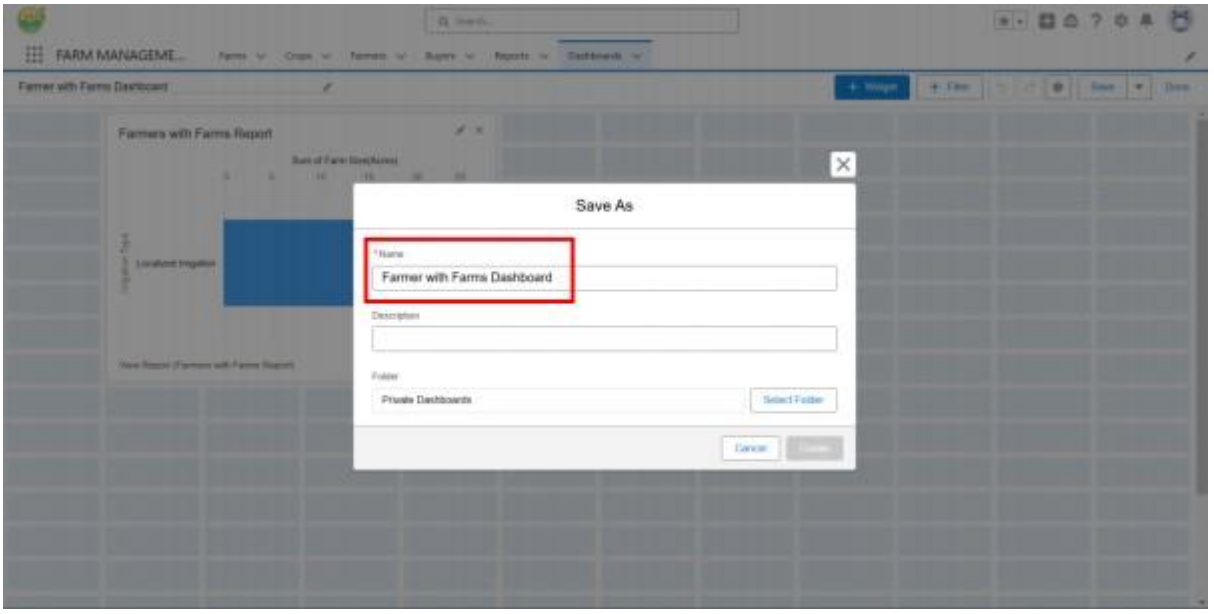
Activity 2:
Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

Milestone 12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:
Create a Dashboard:

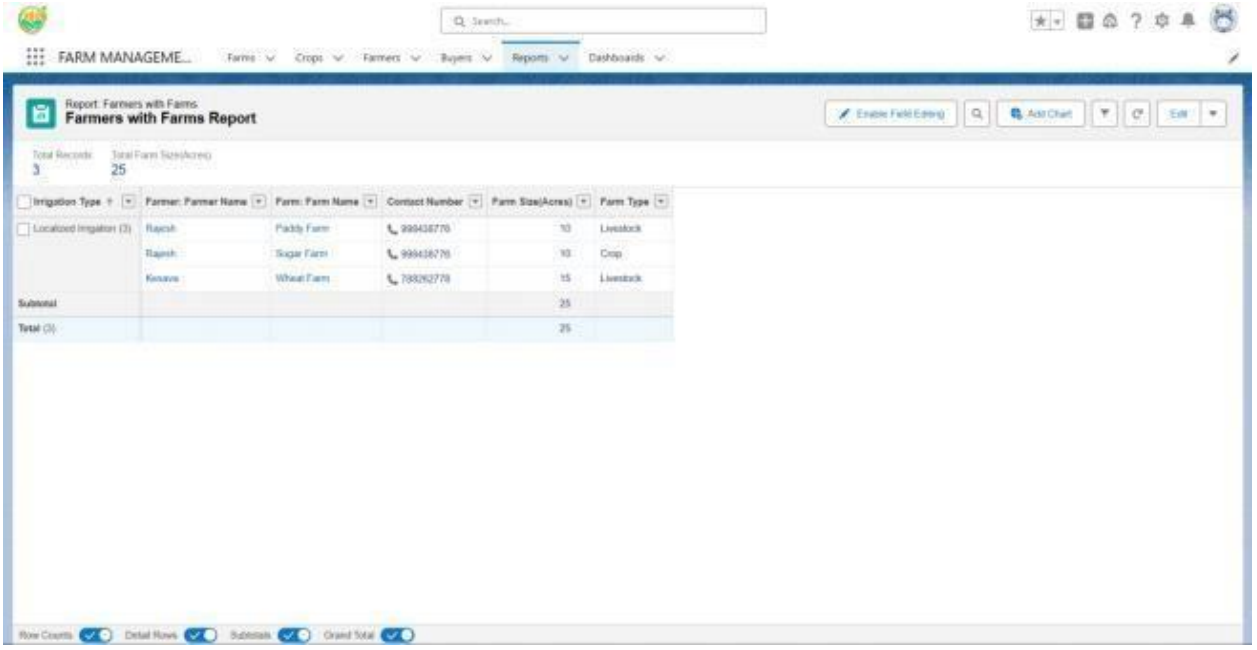
1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.



Milestone-13: View Report and Dashboard:

Activity 1: Report

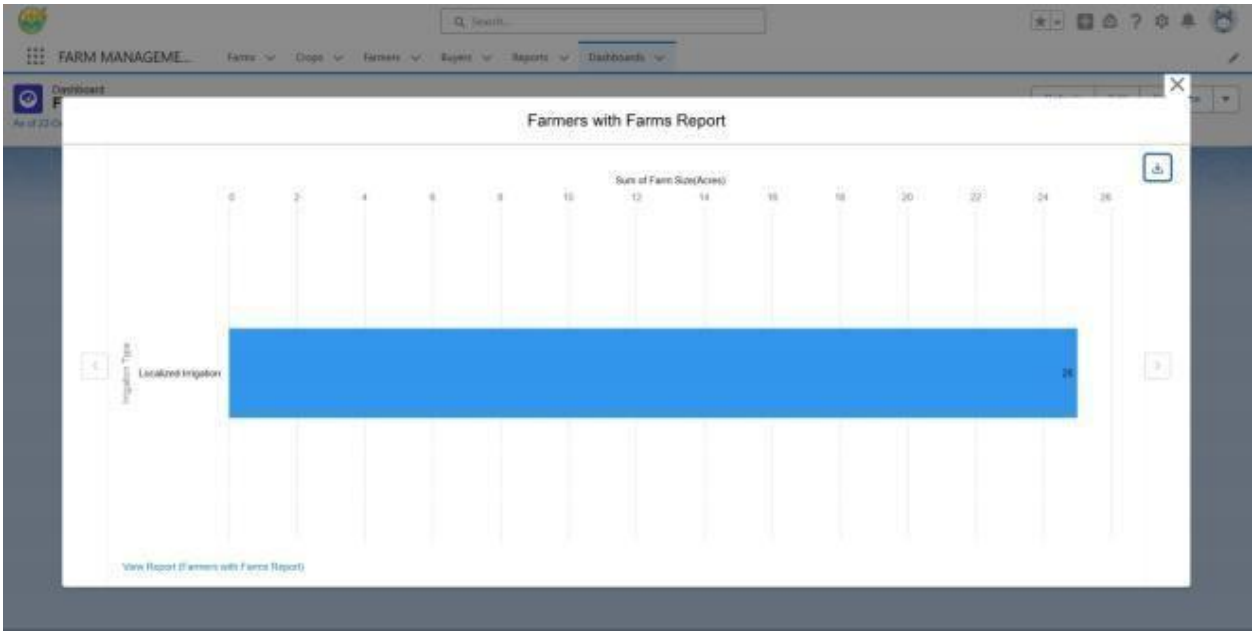
- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records



Activity 2: Dashboard

- Click on App Launcher on left side of screen.

- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.



7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency

- Limited offline access

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURE SCOPE

- Mobile application integration
- AI-based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

11. APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org

Thanking You